

Space Invaders - Specification

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Space Invaders

Menu:

After launching the game, a graphical menu appears where you can navigate using the arrow keys (up-down) to choose between three options: Play, Scoreboard, and Exit.

When selecting Exit, a confirmation window appears asking if you really want to quit. If yes, the game closes; if not, you return to the menu.

Selecting Play opens a second menu where you can choose the difficulty level: Easy, Medium, or Hard.

If Scoreboard is selected in the menu, the top 5 best results are displayed — the record times of players who successfully completed the Hard difficulty.

These results are stored locally in a JSON file.

After selecting a difficulty level, the game starts.

Gameplay:

The player controls a spaceship that moves horizontally and automatically fires a projectile at fixed intervals.

In front of the spaceship, there are alien enemies whose quantity and type depend on the chosen difficulty level.

Like the spaceship, the aliens also shoot projectiles — but towards the spaceship.

They also move horizontally (continuously between the edges of the playing field) and periodically advance closer to the spaceship.

If a projectile from the spaceship hits an alien, it dies (disappears).

If an alien projectile hits the spaceship, the player loses one life point (starting with three).

During the entire game, the elapsed time is displayed in the top-left corner, and the remaining life points are shown in the top-right corner.

End of the Game:

The game can end in three ways:

- The player loses all three life points.
- At least one alien reaches the bottom of the playing field.
- The player completes the selected difficulty level.

To successfully complete the game, the player must defeat all aliens before they reach the bottom of the screen or before losing all three lives.

When the game ends, if the player successfully completes the level, the completion time is displayed.

If the Hard difficulty is completed, the player can enter their name to appear on the scoreboard.

Alien Types:

- Normal: No special abilities; fires at longer intervals.
- Rapid Fire: Fires faster (at shorter intervals) than the normal type.
- Wide: Has a wider appearance than normal and fires three shots at the same rate as the normal type.
- Fast: Moves faster than the normal type and is smaller.
- Fast Rapid Fire: Combines the attributes of the Fast and Rapid Fire types.